Width-Weighted Incidence after the Canonical Affine Quotient: Exact Character Coherence, Sharp Alignment Obstructions, and the Positive-Route Decision

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Abstract

TPC-105 aggregates identical literal affine multiplier maps while retaining their input profiles and complete provenance. This paper analyzes the remaining genuinely distinct-map term. Let $G = \mathbb{F}_q^\times$, let $\phi(j) = A_\phi(m_\phi j + h_0)$ be the canonical maps at one resonance width, and let $f_\phi : G \to [0, \infty)$ be the exact pushforward of the opened-atom input profile. For $M_\phi = \sum_y f_\phi(y)$, the ordered distinct-map excess is

$$\mathfrak{Y}_H^\circ = \sum_{\phi \neq \psi} \left\{ \sum_{y \in G} f_\phi(y) f_\psi(y) - \frac{M_\phi M_\psi}{q-1} \right\}.$$

We prove the exact multiplicative-character identity

$$\mathfrak{Y}_H^\circ = \frac{1}{q-1} \sum_{\chi \neq \chi_0} \sum_{\phi \neq \psi} \widehat{f}_\phi(\chi) \overline{\widehat{f}_\psi(\chi)} = \|a_H^\circ\|_2^2 - \sum_\phi \|f_\phi^\circ\|_2^2, \qquad a_H = \sum_\phi f_\phi.$$

Thus the exact missing datum is cross-map character coherence, not the number of distinct formulas. If $\mathcal{E}_H = \sum_\phi \|f_\phi^\circ\|_2^2$ and $\kappa_H = |\mathfrak{Y}_H^\circ| / \mathcal{E}_H$, then

$$0 \leq \kappa_H \leq |\Phi_H| - 1, \qquad \sum_H g_q(H) \sqrt{|\mathfrak{Y}_H^\circ|} = \sum_H g_q(H) \sqrt{\kappa_H \mathcal{E}_H}.$$

The upper endpoint is sharp even inside the physical map shape $A(mj + h_0)$: distinct maps supported at one input can all hit one output. Hence map distinctness, no-wrap injectivity, and width weighting alone cannot prove the TPC-106 target. We give a two-ledger sufficient criterion and an exact deterministic certificate, but do not estimate the actual growing profiles. The identities and obstruction are L0; their literal crosswalk is L1. No L2 fixed-h_0 arithmetic estimate, parity breakthrough, or prime-pair result is claimed.

Keywords: fixed-shift affine carrier; distinct-map incidence; character coherence; resonance width; positive majorant; parity barrier.

Audit convention. L0 denotes exact finite algebra or a finite countermodel, L1 a literal physical-interface statement, and L2 only a growing fixed nonzero-h_0 arithmetic estimate. The prescribed h_0, physical weights, actual support, and native normalization are never replaced by an averaged or model family.

1 Scope and imported interface

TPC-99–TPC-102 reduce the positive nonconstant resonance sector to a principal ledger, an opened-atom diagonal ledger, and a width-weighted cross-cell ledger [1, 2, 4]. TPC-105 then quotients identical affine maps without replacing an input profile by its total mass [3]. The remaining proposed estimate is

$$\mathfrak{Y}_{\mathrm{dist},X} := \sum_H g_{q_X}(H) \sqrt{|\mathfrak{Y}_H^\circ|} \ll X^{o(1)} Q_X^2, \tag{1}$$

where

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}}.$$

The square root contains an absolute value because the centered ordered cross term need not be nonnegative. Even a proof of (1) closes the full positive cross ledger only together with the independent TPC-105 conditions

$$U_X \le X^{o(1)}, \qquad \mathfrak{P}_X \le X^{o(1)} Q_X^2.$$

Those conditions return the identical-map ledger and are not reproved or assumed true here.

We analyze exactly what information can imply (1). The paper does not assert that the actual TPC profiles satisfy the resulting coherence bounds.

2 Canonical maps and exact pushforwards

Fix an odd prime q, put $G = \mathbb{F}_q^\times$, and fix $h_0 \in G$. At one active width H, let Φ_H be the canonical set of pairwise distinct maps

$$\phi(j) = s_\phi j + t_\phi = A_\phi(m_\phi j + h_0), \qquad A_\phi, m_\phi \in G. \tag{2}$$

The identity with the right-hand expression is part of the literal fixed-h_0 interface.

For each ϕ, let $J_\phi \subseteq \mathbb{F}_q$ be its actual active input support, with $\phi(J_\phi) \subseteq G$, and let $\nu_\phi : J_\phi \to [0, \infty)$ be the provenance-preserving opened-atom profile. Define

$$f_\phi(y) = \sum_{j \in J_\phi} \nu_\phi(j) \mathbf{1}_{\{\phi(j)=y\}}, \tag{3}$$

$$M_\phi = \sum_{j \in J_\phi} \nu_\phi(j) = \sum_{y \in G} f_\phi(y), \qquad a_H = \sum_{\phi \in \Phi_H} f_\phi. \tag{4}$$

Since $s_\phi \neq 0$, each map is injective on $\mathbb{F}_q$; nevertheless different maps may have arbitrarily aligned pushforwards.

Definition 2.1 (Ordered distinct-map excess). Set

$$\mathfrak{Y}_H^\circ = \sum_{\substack{\phi,\psi \in \Phi_H \\ \phi \neq \psi}} \left[\sum_{y \in G} f_\phi(y) f_\psi(y) - \frac{M_\phi M_\psi}{q-1} \right]. \tag{5}$$

The ordered convention agrees with expanding the squared $\ell^2(G)$ norm. Each unordered pair therefore occurs twice.

3 Exact character-coherence theorem

For $f : G \to \mathbb{C}$, use

$$\widehat{f}(\chi) = \sum_{y \in G} f(y) \overline{\chi(y)}, \qquad f^\circ = f - \frac{1}{q-1} \left(\sum_y f(y) \right) \mathbf{1}_G.$$

Theorem 3.1 (Three exact forms of the distinct-map ledger)**.** *For every family in equation (2)–equation (4),*

$$\mathfrak{Y}_H^\circ = \sum_{\phi \neq \psi} \langle f_\phi^\circ, f_\psi^\circ \rangle, \tag{6}$$

$$= \frac{1}{q-1} \sum_{\chi \neq \chi_0} \sum_{\phi \neq \psi} \widehat{f}_\phi(\chi) \overline{\widehat{f}_\psi(\chi)}, \tag{7}$$

$$= \|a_H^\circ\|_2^2 - \sum_{\phi \in \Phi_H} \|f_\phi^\circ\|_2^2. \tag{8}$$

Proof. The bracket in (5) is exactly $\langle f_\phi^\circ, f_\psi^\circ\rangle$, giving (6). Multiplicative Parseval on G deletes the principal character and gives (7). Finally, expand

$$\left\|\sum_\phi f_\phi^\circ\right\|_2^2 = \sum_\phi \|f_\phi^\circ\|_2^2 + \sum_{\phi\neq\psi} \langle f_\phi^\circ, f_\psi^\circ\rangle.$$

□

Remark 3.2 (Why a map total is insufficient). The character vector $\widehat{f}_\phi(\chi)$, not merely $M_\phi = \widehat{f}_\phi(\chi_0)$, determines the nonprincipal cross term. This is the precise reason that the canonical quotient must retain the complete input profile and provenance bag.

Definition 3.3 (Internal energy and coherence). Put

$$\mathcal{E}_H = \sum_{\phi\in\Phi_H} \|f_\phi^\circ\|_2^2, \qquad \kappa_H = \begin{cases} |\mathfrak{Y}_H^\circ|/\mathcal{E}_H, & \mathcal{E}_H > 0, \\ 0, & \mathcal{E}_H = 0. \end{cases}$$

Proposition 3.4 (Sharp universal coherence range)**.** *If* $K_H = |\Phi_H|$*, then*

$$0 \le \kappa_H \le K_H - 1. \tag{9}$$

Proof. The lower bound is definitional. By Cauchy–Schwarz,

$$\begin{aligned} |\mathfrak{Y}_H^\circ| &\le \sum_{\phi\neq\psi} \|f_\phi^\circ\|_2 \|f_\psi^\circ\|_2 \\ &= \left(\sum_\phi \|f_\phi^\circ\|_2\right)^2 - \sum_\phi \|f_\phi^\circ\|_2^2 \le (K_H - 1)\mathcal{E}_H. \end{aligned}$$

□

Corollary 3.5 (Exact two-ledger reduction)**.** *The TPC-106 left side has the exact form*

$$\mathfrak{Y}_{\mathrm{dist},X} = \sum_H g_{q_X}(H)\sqrt{\kappa_H \mathcal{E}_H} \tag{10}$$

and obeys

$$\mathfrak{Y}_{\mathrm{dist},X} \le \left(\sum_H g_{q_X}(H)^2 \kappa_H\right)^{1/2} \left(\sum_H \mathcal{E}_H\right)^{1/2}. \tag{11}$$

In particular, bounds

$$\sum_H g_{q_X}(H)^2 \kappa_H \le X^{o(1)} Q_X^2, \qquad \sum_H \mathcal{E}_H \le X^{o(1)} Q_X^2$$

are sufficient for (1)*.*

Proof. The identity follows from definition 3.3; the inequality is Cauchy–Schwarz in H. □

This reduction separates the amount of centered mass carried by each map from the alignment of distinct maps. Neither ledger is estimated here for the actual growing coefficient.

4 Sharp physical-shape alignment obstruction

The next result shows that the word "distinct" has no quantitative dispersion content by itself.

Theorem 4.1 (Distinct physical maps can align perfectly)**.** *Let $q \geq 5$, choose $h_0, j_0, y_0 \in G$, and let $\mathcal{M} \subseteq G \setminus \{-h_0 j_0^{-1}\}$ have cardinality K. For $m \in \mathcal{M}$, set*

$$A_m = \frac{y_0}{mj_0 + h_0}, \qquad \phi_m(j) = A_m(mj + h_0).$$

Then the ϕ_m are pairwise distinct maps of the literal shape (2). If every profile is the singleton $\nu_m = w\mathbf{1}_{\{j_0\}}$, $w > 0$, then

$$f_m = w\mathbf{1}_{\{y_0\}}, \tag{12}$$

$$\mathcal{E}_H = Kw^2 \frac{q-2}{q-1}, \tag{13}$$

$$\mathfrak{Y}_H^\circ = K(K-1)w^2 \frac{q-2}{q-1}, \tag{14}$$

$$\kappa_H = K - 1. \tag{15}$$

Thus (9) is sharp.

Proof. Every denominator defining A_m is nonzero and $\phi_m(j_0) = y_0$. If $\phi_m = \phi_{m'}$, equality of the constant terms gives $A_m h_0 = A_{m'} h_0$, hence $A_m = A_{m'}$; equality of slopes then gives $m = m'$. The centered norm of $w\mathbf{1}_{\{y_0\}}$ on a set of size $q-1$ is $w^2(1 - 1/(q-1))$. Every ordered distinct pair has the same centered inner product, so the formulas follow. □

Corollary 4.2 (No geometry-only TPC-106 theorem)**.** *No bound with subpower coherence can follow solely from:*

- *pairwise distinctness of the canonical maps;*
- *the fixed-h_0 form $A(mj + h_0)$;*
- *injectivity of each map on its no-wrap input support; and*
- *the resonance width H.*

Additional information about the actual profiles, weights, or arithmetic signs is necessary.

Remark 4.3 (What the countermodel does not prove)*.* The construction is an exact finite obstruction to a coefficient-blind argument. It is not asserted to occur with polynomial mass inside the literal TPC coefficient. Therefore it does not prove that the actual positive route fails.

5 Literal audit and route decision

For the actual coefficient, a proof-carrying TPC-106 certificate must list, at every active H:

(i) every canonical map $\phi(j) = A_\phi(m_\phi j + h_0)$;

(ii) the recoverable provenance bag and exact profile $\nu_\phi(j)$;

(iii) the actual support condition $\phi(j) \neq 0$ and all inherited masks;

(iv) one copy of the physical opened-atom weight under the native global normalization;

(v) the exact values of $\mathcal{E}_H$ and either $\mathfrak{Y}_H^\circ$ or the character vectors in (7); and

(vi) the width-weighted sum in (10).

Ambient map counts or a random-map model cannot replace these data.

Proposition 5.1 (Stop–go statement)**.** *The following conclusions have different strengths.*

(a) A verified bound (1) *for the actual growing profiles is a go certificate for the distinct-map part of the positive resonance route.*

(b) A verified polynomial lower obstruction in the actual $\mathfrak{Y}_{\mathrm{dist},X}$*, at its native normalization and with no compensating signed term, stops this positive termwise route and triggers signed filtering before majorization.*

(c) Corollary 4.2 already stops every proof based only on map distinctness, but does not decide (a) or (b).

6 Deterministic regression certificate

The script `experiments/tpc106_certificate.py` checks, over several small primes:

- equality of the direct centered-Gram form and the energy-difference form in theorem 3.1;
- the sharp family in theorem 4.1; and
- the coherence bound $\kappa_H \le K_H - 1$.

It uses exact rational arithmetic. The character form in theorem 3.1 is proved analytically by multiplicative Fourier inversion; it is not separately tested by this script. The certificate verifies finite algebra, not asymptotic arithmetic cancellation.

7 Claim boundary

Proved

The canonical distinct-map ledger has the exact character and Gram forms in theorem 3.1. Its width-weighted target is exactly the energy–coherence ledger (10). The universal coherence factor is at most $K_H - 1$, and this endpoint is achieved by pairwise distinct maps of the literal fixed-h_0 shape. These are L0 results. Importing the TPC-105 quotient profiles without changing support, coefficients, or normalization gives the stated L1 interface.

Not proved

This paper does not prove:

- (1) for the actual growing profiles;
- a subpower bound for $\mathcal{E}_H$ or κ_H;
- that the sharp alignment countermodel has substantial mass in the physical packet;
- closure or failure of the full positive resonance route;
- a signed affine Möbius estimate, determinant/zero-mode compatibility, full-block return, or the strict physical endpoint; or
- a parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

Nothing is specialized to $h_0 = 2$.

Verdict

TPC-106 makes a genuine route decision at the theorem-design level: canonical quotienting removes artificial identical-map duplication, but distinctness alone supplies no dispersion. The remaining positive door is precisely a weighted character-coherence theorem for the actual profiles. If that theorem cannot be proved or is falsified on the literal coefficient, the mathematically correct fallback is to retain the signs before taking absolute values, as developed in TPC-107.

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